Large-Scale Agentic Web Experiment

Decisive stress test for *3D Lineage-First Mycelial Fabric — Continual Discovery*.

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Scale

Agents	100, 1,000, 10,000, 100,000
Questions	127,374 labelled evaluation questions per seed at N = 10,000; 17,324,700 question instances answered per system across the main runs
Trials (seeds)	30 at N ≤ 10,000; 10 at N = 100,000; 15 per ablation sweep
Failures simulated	6 failure regimes over 31 regime×severity cells, 3 corruption models × 4 levels, 2 partition topologies × 2 phases = **47 conditions per seed** at N = 10,000
Systems	8 primary (B0–B7) plus 4 ablation cells (B3L, B6C, B2L, B3Q)
Total experimental conditions	50,700 recorded (seed × condition × system) rows across 39 runs
Runtime	97.2 CPU-minutes total, one machine, no GPU, **no LLM calls**

> **Computational cost is reported separately and is entirely simulation.** There are no model calls in this experiment: semantic reasoning is abstracted to symbolic value matching (see *Limitations*).

Primary result

Across **30 seeds at N = 10,000**, under a **white-box targeted lineage attack** that removes 50% of agents — the attacker knows each system's own placement and greedily destroys the hosts of the most important surviving independent evidence — the lineage-aware fabric (B6) recovered **0.679** of valid knowledge versus **0.678** for random replication (B3) and **0.677** for the diversity-blind replica-count policy (B5), all three at the identical storage budget of 4.0 published replicas per claim.

On raw knowledge survival the equal-budget policies are close: B6 – B3 Δ = +0.000 (95% CI [-0.001, 0.001], d_z = 0.10, paired t p = 0.59, Holm p = 0.59, n = 30). The separation appears in the quantities the architecture is actually about:

- **task accuracy**: B6 – B3 Δ = +0.010 (95% CI [0.009, 0.012], d_z = 2.29, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- **independent-support survival (ISS)**: B6 – B3 Δ = +0.046 (95% CI [0.046, 0.047], d_z = 21.30, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- **contradiction detection F1**: B6 – B3 Δ = +0.155 (95% CI [0.149, 0.161], d_z = 8.90, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- **contradiction resolution** (accuracy on claims whose evidence genuinely disagrees): B6 – B3 Δ = +0.003 (95% CI [-0.001, 0.008], d_z = 0.26, paired t p = 0.16, Holm p = 0.33, n = 30)

Broad replication (B2) reached **0.870** survival under the same attack, but at 193.7 replicas per claim — **48× the storage** of the lineage fabric. The centralized store (B1) reached 0.876 at 36.3 replicas per claim, and is included as a performance upper baseline, not as a decentralised competitor.

Table 1 — main results

Mean over seeds; 95% t-intervals in `tables/table1_main_results.csv`. KS = knowledge survival (recoverable **and correct**), ISS = independent-support survival, storage = published replicas per claim, messages = total messages per claim (publication + questioning + probing).

condition	sys	KS	acc	ISS	contr. F1	contr. resolved	ECE*	storage/claim	msgs/claim
50% random churn	B0	0.219	0.081	0.142	0.000	0.127	0.007	0.000	0.000
50% random churn	B1	0.730	0.499	0.833	0.781	0.426	0.007	36.328	42.475
50% random churn	B2	0.876	0.605	1.000	0.900	0.513	0.006	193.749	209.462

condition	sys	KS	acc	ISS	contr. F1	contr. resolved	ECE*	storage/claim	msgs/claim
50% random churn	B3	0.820	0.562	0.622	0.459	0.474	0.006	4.000	8.000
50% random churn	B4	0.864	0.586	0.960	0.896	0.512	0.007	48.437	99.588
50% random churn	B5	0.820	0.551	0.530	0.000	0.468	0.004	4.000	8.000
50% random churn	B6	0.821	0.574	0.686	0.638	0.477	0.006	4.000	8.000
50% random churn	B7	0.881	0.763	0.686	0.486	0.630	0.004	4.000	14.517
50% random churn	B3Q	0.899	0.807	0.628	0.298	0.699	0.003	4.000	15.681
90% random churn	B0	0.044	0.018	0.028	0.000	0.026	0.019	0.000	0.000
90% random churn	B1	0.204	0.140	0.233	0.218	0.118	0.006	36.328	46.769
90% random churn	B2	0.843	0.575	0.895	0.868	0.509	0.007	193.749	251.032
90% random churn	B3	0.301	0.174	0.200	0.036	0.173	0.009	4.000	8.000
90% random churn	B4	0.669	0.431	0.528	0.639	0.460	0.007	48.437	118.795
90% random churn	B5	0.301	0.169	0.195	0.000	0.171	0.007	4.000	8.000
90% random churn	B6	0.300	0.177	0.203	0.058	0.172	0.008	4.000	8.000
90% random churn	B7	0.322	0.237	0.203	0.040	0.226	0.004	4.000	14.517
90% random churn	B3Q	0.328	0.251	0.200	0.024	0.252	0.005	4.000	15.681
50% loss, whole organisations	B0	0.221	0.083	0.142	0.000	0.126	0.011	0.000	0.000
50% loss, whole organisations	B1	0.789	0.539	0.900	0.843	0.460	0.007	36.328	41.998
50% loss, whole organisations	B2	0.877	0.605	1.000	0.899	0.515	0.006	193.749	209.472
50% loss, whole organisations	B3	0.820	0.561	0.622	0.462	0.473	0.005	4.000	8.000
50% loss, whole organisations	B4	0.787	0.494	0.750	0.772	0.505	0.008	48.437	102.565
50% loss, whole organisations	B5	0.820	0.550	0.530	0.000	0.469	0.003	4.000	8.000
50% loss, whole organisations	B6	0.822	0.575	0.687	0.639	0.477	0.006	4.000	8.000
50% loss, whole organisations	B7	0.883	0.764	0.687	0.489	0.633	0.004	4.000	14.517
50% loss, whole organisations	B3Q	0.898	0.806	0.628	0.298	0.701	0.004	4.000	15.681
70% loss, whole failure domains	B0	0.132	0.050	0.085	0.000	0.076	0.010	0.000	0.000
70% loss, whole failure domains	B1	0.672	0.460	0.767	0.719	0.392	0.007	36.328	42.955
70% loss, whole failure domains	B2	0.876	0.604	0.998	0.899	0.512	0.007	193.749	218.316
70% loss, whole failure domains	B3	0.665	0.431	0.471	0.238	0.385	0.005	4.000	8.000
70% loss, whole failure domains	B4	0.763	0.490	0.696	0.747	0.494	0.007	48.437	107.307
70% loss, whole failure domains	B5	0.665	0.421	0.430	0.000	0.381	0.005	4.000	8.000
70% loss, whole failure domains	B6	0.667	0.442	0.499	0.359	0.387	0.007	4.000	8.000
70% loss, whole failure domains	B7	0.716	0.586	0.499	0.262	0.506	0.004	4.000	14.517
70% loss, whole failure domains	B3Q	0.727	0.617	0.473	0.151	0.564	0.004	4.000	15.681
50% targeted lineage attack	B0	0.111	0.041	0.067	0.000	0.069	0.014	0.000	0.000
50% targeted lineage attack	B1	0.876	0.600	1.000	0.937	0.511	0.007	36.328	41.286
50% targeted lineage attack	B2	0.870	0.597	0.993	0.895	0.513	0.006	193.749	210.089
50% targeted lineage attack	B3	0.678	0.441	0.504	0.351	0.390	0.006	4.000	8.000
50% targeted lineage attack	B4	0.749	0.472	0.691	0.747	0.492	0.007	48.437	105.090
50% targeted lineage attack	B5	0.677	0.429	0.437	0.000	0.385	0.005	4.000	8.000
50% targeted lineage attack	B6	0.679	0.451	0.550	0.506	0.394	0.008	4.000	8.000
50% targeted lineage attack	B7	0.726	0.594	0.550	0.398	0.511	0.004	4.000	14.517
50% targeted lineage attack	B3Q	0.740	0.628	0.507	0.243	0.566	0.004	4.000	15.681
20% origin-source corruption	B0	0.246	0.090	0.198	0.000	0.197	0.012	0.000	0.000
20% origin-source corruption	B1	0.675	0.431	0.933	0.870	0.577	0.010	36.328	41.763
20% origin-source corruption	B2	0.719	0.464	1.000	0.894	0.608	0.010	193.749	205.168
20% origin-source corruption	B3	0.702	0.456	0.706	0.628	0.574	0.008	4.000	8.000
20% origin-source corruption	B4	0.719	0.461	0.994	0.901	0.611	0.009	48.437	96.845
20% origin-source corruption	B5	0.695	0.444	0.561	0.000	0.554	0.009	4.000	8.000
20% origin-source corruption	B6	0.717	0.475	0.810	0.827	0.611	0.009	4.000	8.000
20% origin-source corruption	B7	0.770	0.640	0.810	0.794	0.660	0.009	4.000	14.517
20% origin-source corruption	B3Q	0.770	0.667	0.715	0.610	0.642	0.008	4.000	15.681
20% coordinated node corruption	B0	0.245	0.090	0.199	0.000	0.223	0.012	0.000	0.000
20% coordinated node corruption	B1	0.672	0.459	0.967	0.206	0.663	0.006	36.328	41.520
20% coordinated node corruption	B2	0.848	0.578	1.000	0.722	0.834	0.007	193.749	205.169

condition	sys	KS	acc	ISS	contr. F1	contr. resolved	ECE*	storage/claim	msgs/claim
20% coordinated node corruption	B3	0.751	0.498	0.706	0.540	0.739	0.009	4.000	8.000
20% coordinated node corruption	B4	0.821	0.539	0.994	0.764	0.825	0.009	48.437	96.841
20% coordinated node corruption	B5	0.752	0.492	0.561	0.504	0.741	0.009	4.000	8.000
20% coordinated node corruption	B6	0.750	0.507	0.810	0.557	0.738	0.010	4.000	8.000
20% coordinated node corruption	B7	0.733	0.589	0.810	0.576	0.706	0.009	4.000	14.517
20% coordinated node corruption	B3Q	0.722	0.600	0.714	0.555	0.689	0.011	4.000	15.681
4-way partition, after reconnect	B0	0.392	0.191	0.283	0.000	0.240	0.008	0.000	0.000
4-way partition, after reconnect	B1	0.672	0.363	1.000	0.421	0.297	0.006	36.328	41.286
4-way partition, after reconnect	B2	0.659	0.357	1.000	0.797	0.259	0.008	193.749	201.749
4-way partition, after reconnect	B3	0.677	0.375	0.773	0.701	0.305	0.008	4.000	8.000
4-way partition, after reconnect	B4	0.759	0.431	1.000	0.850	0.422	0.008	48.437	94.467
4-way partition, after reconnect	B5	0.676	0.368	0.566	0.561	0.304	0.006	4.000	8.000
4-way partition, after reconnect	B6	0.674	0.382	0.930	0.761	0.299	0.012	4.000	8.000
4-way partition, after reconnect	B7	0.829	0.695	0.930	0.622	0.548	0.007	4.000	15.393
4-way partition, after reconnect	B3Q	0.882	0.792	0.786	0.463	0.666	0.005	4.000	16.881

ECE is calibration error after held-out histogram recalibration; raw ECE and confidence AUROC are in `tables/appendix_full_grid.csv`.*

Continual questioning result

What B3+Q controls for. B3+Q applies the identical questioning machinery to random placement with replica-count aggregation. It is not a *lineage-free* system: re-verifying a stored replica requires knowing which origin produced it, so provenance metadata is still present. What B3+Q removes is lineage-aware *placement* (choosing which evidence to publish so that distinct origins are covered) and lineage-aware *aggregation* (counting distinct origins rather than replicas). The comparison therefore answers: does continual discovery need those two mechanisms, or only the provenance pointer it uses to ask?

G_Q is measured **at identical storage**: the questioning system's base placement budget is reduced by exactly the number of replicas its questions add, so the only extra cost is messages. Paired by seed, n = 30.

condition	metric	B6	B7	G_Q	d_z	Holm p	G_Q on B3 base
50% loss, whole organisations	accuracy_macro	0.575	0.764	0.189	37.157	0.000	0.245
50% loss, whole organisations	acc_revision_sensitive	0.142	0.593	0.451	44.437	0.000	0.593
50% loss, whole organisations	knowledge_survival	0.822	0.883	0.061	18.591	0.000	0.078
50% loss, whole organisations	contradiction_f1	0.639	0.489	-0.150	-11.949	0.000	-0.163
50% loss, whole organisations	contradiction_resolved_acc	0.477	0.633	0.155	12.246	0.000	0.229
50% targeted lineage attack	accuracy_macro	0.451	0.594	0.143	32.914	0.000	0.187
50% targeted lineage attack	acc_revision_sensitive	0.120	0.469	0.349	48.481	0.000	0.467
50% targeted lineage attack	knowledge_survival	0.679	0.726	0.047	16.405	0.000	0.061
50% targeted lineage attack	contradiction_f1	0.506	0.398	-0.108	-5.507	0.000	-0.109
50% targeted lineage attack	contradiction_resolved_acc	0.394	0.511	0.118	9.541	0.000	0.175
50% random churn	accuracy_macro	0.574	0.763	0.189	44.472	0.000	0.246
50% random churn	acc_revision_sensitive	0.142	0.590	0.447	54.517	0.000	0.593
50% random churn	knowledge_survival	0.821	0.881	0.060	23.198	0.000	0.078
50% random churn	contradiction_f1	0.638	0.486	-0.152	-9.171	0.000	-0.161
50% random churn	contradiction_resolved_acc	0.477	0.630	0.153	12.793	0.000	0.225
20% origin-source corruption	accuracy_macro	0.475	0.640	0.165	29.336	0.000	0.210
20% origin-source corruption	acc_revision_sensitive	0.117	0.511	0.395	29.488	0.000	0.519
20% origin-source corruption	knowledge_survival	0.717	0.770	0.054	15.508	0.000	0.068
20% origin-source corruption	contradiction_f1	0.827	0.794	-0.033	-8.154	0.000	-0.018
20% origin-source corruption	contradiction_resolved_acc	0.611	0.660	0.049	6.494	0.000	0.069
20% coordinated node corruption	accuracy_macro	0.507	0.589	0.082	10.830	0.000	0.102
20% coordinated node corruption	acc_revision_sensitive	0.117	0.438	0.321	29.604	0.000	0.426
20% coordinated node corruption	knowledge_survival	0.750	0.733	-0.017	-3.192	0.000	-0.029
20% coordinated node corruption	contradiction_f1	0.557	0.576	0.018	3.156	0.000	0.015
20% coordinated node corruption	contradiction_resolved_acc	0.738	0.706	-0.031	-6.556	0.000	-0.050

Note the two contradiction metrics move in opposite directions for the questioning system, and that is the mechanism working as designed: re-verification *removes* the disagreement from the fabric, so there is less disagreement left to detect (F1 falls) while more contradictory claims end up answered correctly (resolution rises).

The cost of that gain: 14.52 messages per claim versus 8.00 for B6 without questioning — a 1.81× increase in communication for zero increase in storage.

Question budget

sys	questions/claim	accuracy	stale-fact acc	contr. F1	msgs/claim
B6	0.050	0.576	0.142	0.640	8.000
B6	0.100	0.576	0.142	0.640	8.000
B6	0.200	0.576	0.142	0.640	8.000
B6	0.400	0.576	0.142	0.640	8.000
B6	0.600	0.576	0.142	0.640	8.000
B6	1.000	0.576	0.142	0.640	8.000
B6	2.000	0.576	0.142	0.640	8.000
B7	0.050	0.601	0.208	0.635	8.531
B7	0.100	0.622	0.259	0.631	9.064
B7	0.200	0.659	0.350	0.618	10.127
B7	0.400	0.713	0.480	0.572	12.255
B7	0.600	0.765	0.593	0.492	14.516
B7	1.000	0.850	0.780	0.276	19.356
B7	2.000	0.897	0.882	0.141	26.399

Question-targeting strategy

targeting	accuracy	stale-fact acc	KS	ISS	contr. F1
hybrid	0.765	0.593	0.884	0.688	0.492
random	0.720	0.475	0.872	0.688	0.429
strategic	0.719	0.486	0.869	0.688	0.517
temporal	0.773	0.608	0.889	0.688	0.421

Which question type does the work

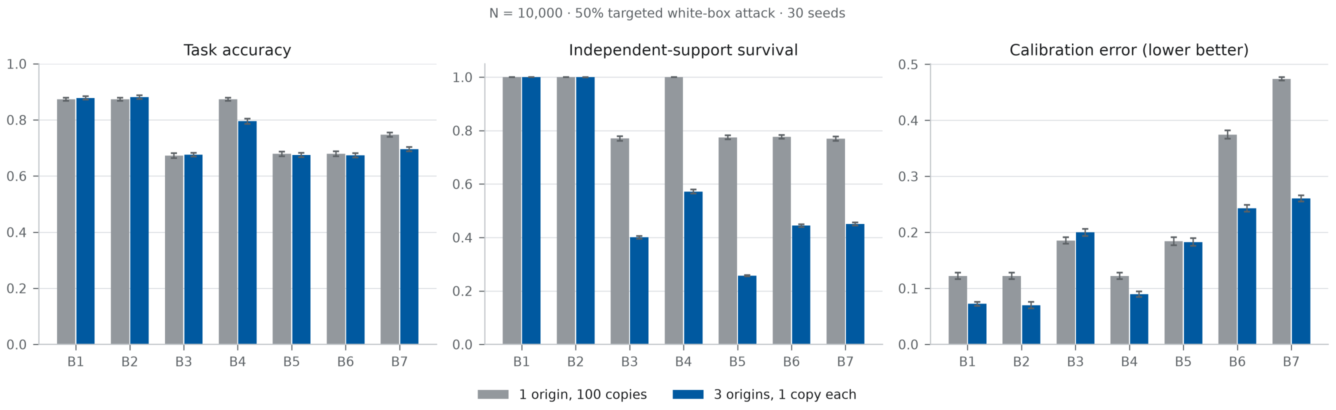
question type	accuracy	stale-fact acc	KS	ISS	contr. F1	msgs/claim
RESOLVE only	0.818	0.713	0.902	0.688	0.406	17.120
ADD_SUPPORT only	0.602	0.200	0.833	0.688	0.598	12.090
REVERIFY only	0.817	0.714	0.901	0.687	0.405	14.718

Correlated replication result

The world contains two designated claim classes that isolate the core distinction. A *correlated* claim has **one** origin with **100** derived copies (`e0` → `e1..e100`); an *independent* claim has **three** origins with one copy each. Both are measured under the same 50% white-box targeted attack.

sys	acc (1 origin ×100)	acc (3 origins ×1)	acc gap	ISS corr.	ISS indep.	ECE corr.	ECE indep.
B1	0.874	0.878	0.005	1.000	1.000	0.122	0.072
B2	0.874	0.881	0.008	1.000	1.000	0.122	0.070
B3	0.673	0.676	0.003	0.770	0.401	0.186	0.200
B4	0.874	0.795	-0.079	1.000	0.572	0.122	0.090
B5	0.679	0.675	-0.004	0.775	0.257	0.184	0.183
B6	0.679	0.674	-0.006	0.777	0.445	0.375	0.243
B7	0.748	0.696	-0.051	0.770	0.451	0.474	0.261

Reading the table: three independent origins are worth less than one hundred correlated copies for every system (accuracy gap -0.006 for B6, -0.004 for B5, +0.008 for B2). The replica-count policy B5, which keeps the *best-corroborated* evidence, is the one that most systematically confuses the two: its independent-support survival on correlated claims is 0.775 against 0.777 for the lineage fabric.



correlated vs independent

Scaling result

Measured at 50% random churn across every scale that was run. **These are empirical scaling trends from a handful of measured points, not asymptotic complexity claims.**

sys	quantity	points	log-log slope	R^2	trend	measured (N:value)
B1	storage_per_claim	4	0.001	0.909	approximately constant	100:36.02;1000:36.1;10000:36.33;100000:36.35
B1	messages_per_claim	4	0.001	0.081	approximately constant	100:41.92;1000:42.23;10000:42.48;100000:42.02
B1	mean_probe_rounds	4	-0.005	0.166	approximately constant	100:1.785;1000:1.849;10000:1.849;100000:1.722
B1	eval_seconds	4	0.643	0.868	sub-linear (near sqrt)	100:0.004047;1000:0.003814;10000:0.02589;100000:0.2963
B1	knowledge_survival	4	0.005	0.175	varies with N	100:0.7579;1000:0.7301;10000:0.7301;100000:0.7885
B2	storage_per_claim	4	0.001	0.909	approximately constant	100:192.1;1000:192.5;10000:193.7;100000:193.9
B2	messages_per_claim	4	0.001	0.915	approximately constant	100:207.8;1000:208.3;10000:209.5;100000:209.6
B2	mean_probe_rounds	4	0.000	0.508	approximately constant	100:4.263;1000:4.274;10000:4.271;100000:4.273
B2	eval_seconds	4	0.810	0.867	sub-linear (near sqrt)	100:0.01244;1000:0.01256;10000:0.1174;100000:2.963
B2	knowledge_survival	4	0.000	0.627	flat in N	100:0.8738;1000:0.8764;10000:0.8763;100000:0.8765
B3	storage_per_claim	4	0.000	--	approximately constant	100:4;1000:4;10000:4;100000:4
B3	messages_per_claim	4	0.000	--	approximately constant	100:8;1000:8;10000:8;100000:8
B3	mean_probe_rounds	4	0.000	--	approximately constant	100:1;1000:1;10000:1;100000:1
B3	eval_seconds	4	0.615	0.874	sub-linear (near sqrt)	100:0.004087;1000:0.004059;10000:0.02454;100000:0.2516
B3	knowledge_survival	4	0.000	0.924	flat in N	100:0.8192;1000:0.8202;10000:0.8205;100000:0.821
B4	storage_per_claim	4	0.001	0.909	approximately constant	100:48.02;1000:48.13;10000:48.44;100000:48.46
B4	messages_per_claim	4	0.001	0.896	approximately constant	100:98.88;1000:99.01;10000:99.59;100000:99.64
B4	mean_probe_rounds	4	-0.000	0.068	approximately constant	100:3.322;1000:3.307;10000:3.313;100000:3.316
B4	eval_seconds	4	0.699	0.874	sub-linear (near sqrt)	100:0.006262;1000:0.006175;10000:0.04822;100000:0.6756
B4	knowledge_survival	4	0.001	0.608	flat in N	100:0.8587;1000:0.8641;10000:0.864;100000:0.8642
B6	storage_per_claim	4	0.000	--	approximately constant	100:4;1000:4;10000:4;100000:4
B6	messages_per_claim	4	0.000	--	approximately constant	100:8;1000:8;10000:8;100000:8
B6	mean_probe_rounds	4	0.000	--	approximately constant	100:1;1000:1;10000:1;100000:1
B6	eval_seconds	4	0.629	0.871	sub-linear (near sqrt)	100:0.003912;1000:0.003848;10000:0.0239;100000:0.2666
B6	knowledge_survival	4	-0.000	0.082	flat in N	100:0.8218;1000:0.8204;10000:0.8209;100000:0.8212
B7	storage_per_claim	4	0.000	--	approximately constant	100:4;1000:4;10000:4;100000:4
B7	messages_per_claim	4	0.000	0.975	approximately constant	100:14.51;1000:14.51;10000:14.52;100000:14.52
B7	mean_probe_rounds	4	0.000	--	approximately constant	100:1;1000:1;10000:1;100000:1
B7	eval_seconds	4	0.626	0.875	sub-linear (near sqrt)	100:0.003885;1000:0.003773;10000:0.02487;100000:0.2525
B7	knowledge_survival	4	-0.000	0.195	flat in N	100:0.8826;1000:0.8808;10000:0.8813;100000:0.8816

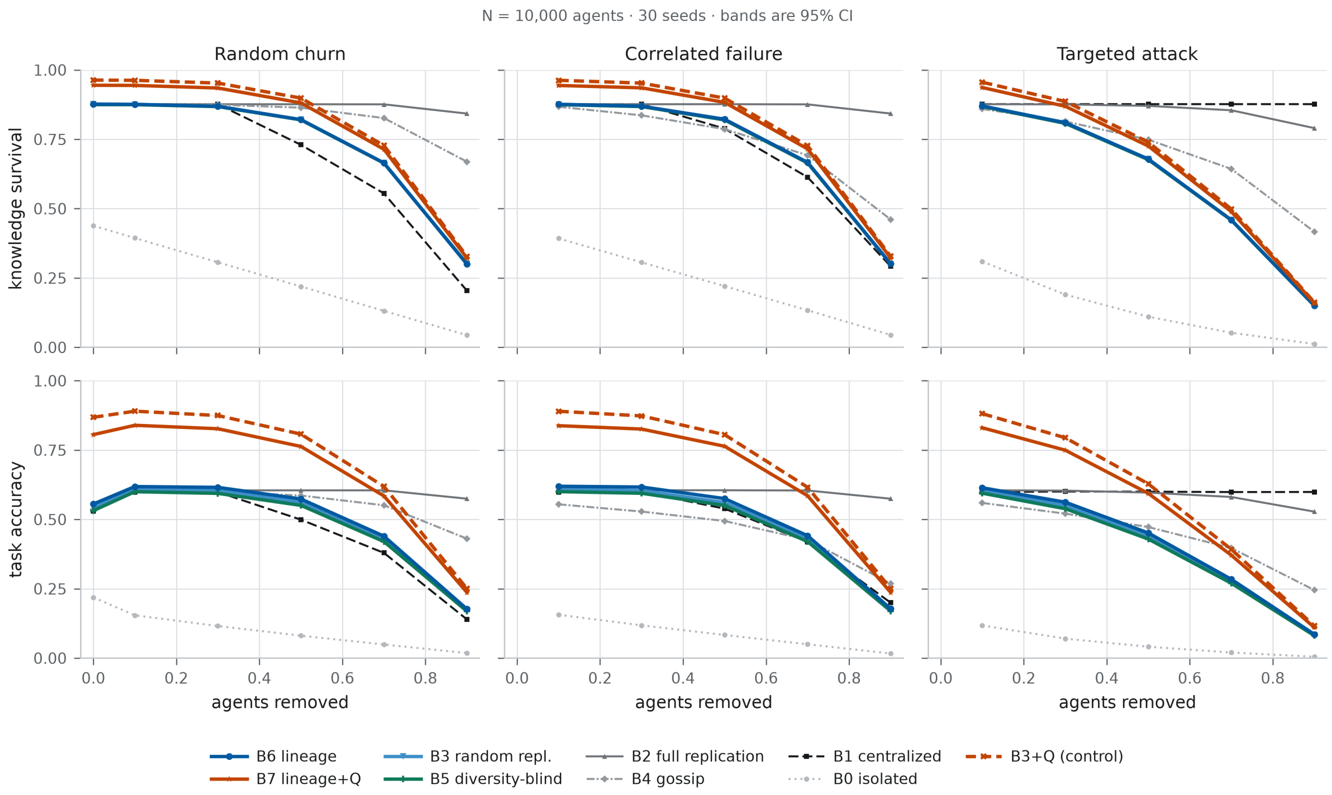
scaling

For the lineage fabric, per-claim storage is fixed by the budget k and therefore does not grow with N (4.00 at N = 100 → 4.00 at N = 100,000); retrieval cost is 1.00 → 1.00 rounds, and knowledge survival is 0.822 → 0.821. The behaviour therefore persists at N ≥ 10■; the total knowledge base grows with N (2 claims per agent), so per-claim costs that stay flat mean total system cost grows linearly in N rather than super-linearly.

Failure-domain result

Correlated failure removes whole failure domains, organisations or regions, trimmed to exactly the same node-level severity as the random regime, so the comparison isolates *structure* of the failure rather than its size.

condition	system	KS	accuracy	ISS
50% random churn	B1	0.730	0.499	0.833
50% random churn	B2	0.876	0.605	1.000
50% random churn	B3	0.820	0.562	0.622
50% random churn	B4	0.864	0.586	0.960
50% random churn	B5	0.820	0.551	0.530
50% random churn	B6	0.821	0.574	0.686
50% random churn	B7	0.881	0.763	0.686
50% loss, whole organisations	B1	0.789	0.539	0.900
50% loss, whole organisations	B2	0.877	0.605	1.000
50% loss, whole organisations	B3	0.820	0.561	0.622
50% loss, whole organisations	B4	0.787	0.494	0.750
50% loss, whole organisations	B5	0.820	0.550	0.530
50% loss, whole organisations	B6	0.822	0.575	0.687
50% loss, whole organisations	B7	0.883	0.764	0.687
70% loss, whole failure domains	B1	0.672	0.460	0.767
70% loss, whole failure domains	B2	0.876	0.604	0.998
70% loss, whole failure domains	B3	0.665	0.431	0.471
70% loss, whole failure domains	B4	0.763	0.490	0.696
70% loss, whole failure domains	B5	0.665	0.421	0.430
70% loss, whole failure domains	B6	0.667	0.442	0.499
70% loss, whole failure domains	B7	0.716	0.586	0.499
50% targeted lineage attack	B1	0.876	0.600	1.000
50% targeted lineage attack	B2	0.870	0.597	0.993
50% targeted lineage attack	B3	0.678	0.441	0.504
50% targeted lineage attack	B4	0.749	0.472	0.691
50% targeted lineage attack	B5	0.677	0.429	0.437
50% targeted lineage attack	B6	0.679	0.451	0.550
50% targeted lineage attack	B7	0.726	0.594	0.550
90% random churn	B1	0.204	0.140	0.233
90% random churn	B2	0.843	0.575	0.895
90% random churn	B3	0.301	0.174	0.200
90% random churn	B4	0.669	0.431	0.528
90% random churn	B5	0.301	0.169	0.195
90% random churn	B6	0.300	0.177	0.203
90% random churn	B7	0.322	0.237	0.203



money figure

Partition and reconciliation result

The network is split into 2 or 4 components; facts are updated independently inside the component that holds their origin; then the network reconnects. `acc_updated` is the reconciliation rate — of the claims updated during the split, the fraction that carry the post-update value afterwards. `conflict_visible` is the fraction of those claims where the system can actually *see* the disagreement in the evidence it retrieves.

sys	KS	accuracy	reconciliation rate	conflict visible	contr. F1
B0	0.392	0.191	0.284	0.000	0.000
B1	0.672	0.363	0.196	0.021	0.421
B2	0.659	0.357	0.153	0.804	0.797
B3	0.677	0.375	0.211	0.626	0.701
B4	0.759	0.431	0.485	0.761	0.850
B5	0.676	0.368	0.214	0.618	0.561
B6	0.674	0.382	0.203	0.631	0.761
B7	0.829	0.695	0.554	0.368	0.622
B3Q	0.882	0.792	0.688	0.262	0.463

- After reconnection, B6 – B3 accuracy Δ = +0.007 (95% CI [0.005, 0.008], d_z = 1.72, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- After reconnection, B7 – B6 accuracy Δ = +0.314 (95% CI [0.311, 0.316], d_z = 49.75, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- After reconnection, B7 – B3Q accuracy Δ = -0.096 (95% CI [-0.098, -0.095], d_z = -19.26, paired t p = <1e-6, Holm p = <1e-6, n = 30)

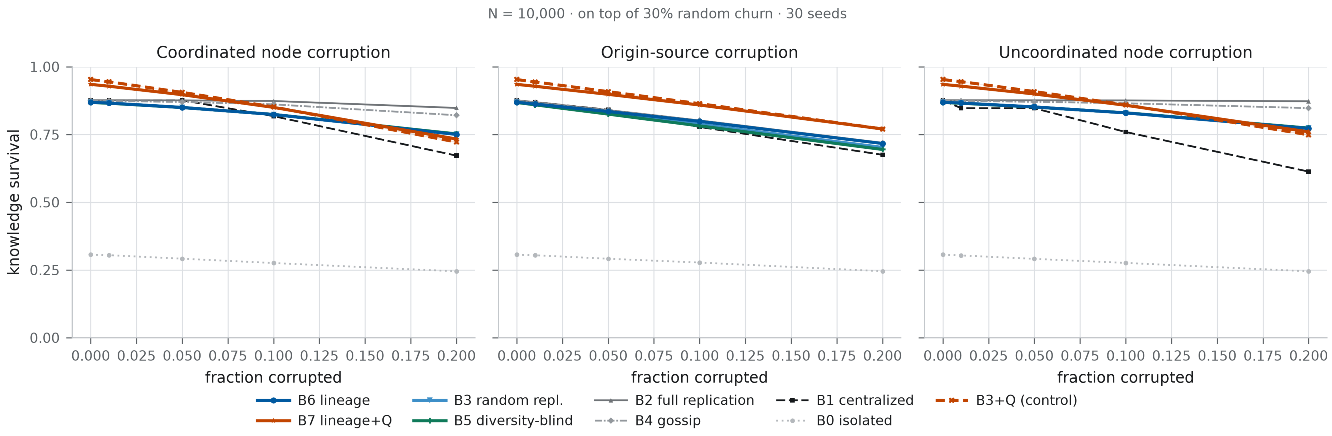
Corruption result

Three corruption models are injected on top of 30% random churn: **coordinated node corruption** (all corrupt agents assert the *same* false value — the worst case for majority voting), **uncoordinated node corruption**, and **origin-source**

corruption, in which a corrupted origin's false value is inherited by every one of its derived copies. The last model is the one that separates replica count from independent support.

condition	system	KS	accuracy	ISS	contradiction_F1
20% coordinated node corruption	B1	0.672	0.459	0.967	0.206
20% coordinated node corruption	B2	0.848	0.578	1.000	0.722
20% coordinated node corruption	B3	0.751	0.498	0.706	0.540
20% coordinated node corruption	B4	0.821	0.539	0.994	0.764
20% coordinated node corruption	B5	0.752	0.492	0.561	0.504
20% coordinated node corruption	B6	0.750	0.507	0.810	0.557
20% coordinated node corruption	B7	0.733	0.589	0.810	0.576
20% origin-source corruption	B1	0.675	0.431	0.933	0.870
20% origin-source corruption	B2	0.719	0.464	1.000	0.894
20% origin-source corruption	B3	0.702	0.456	0.706	0.628
20% origin-source corruption	B4	0.719	0.461	0.994	0.901
20% origin-source corruption	B5	0.695	0.444	0.561	0.000
20% origin-source corruption	B6	0.717	0.475	0.810	0.827
20% origin-source corruption	B7	0.770	0.640	0.810	0.794

- Under 20% origin-source corruption, B6 – B3 accuracy Δ = +0.019 (95% CI [0.018, 0.020], d_z = 5.14, paired t p = <1e-6, Holm p = <1e-6, n = 30); contradiction F1 Δ = +0.199 (95% CI [0.197, 0.201], d_z = 37.05, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- Under 20% origin-source corruption, B6 – B5 accuracy Δ = +0.031 (95% CI [0.029, 0.032], d_z = 7.14, paired t p = <1e-6, Holm p = <1e-6, n = 30); contradiction F1 Δ = +0.827 (95% CI [0.826, 0.828], d_z = 266.03, paired t p = <1e-6, Holm p = <1e-6, n = 30)
- Under 20% origin-source corruption, B6 – B2 accuracy Δ = +0.011 (95% CI [0.010, 0.013], d_z = 2.87, paired t p = <1e-6, Holm p = <1e-6, n = 30); contradiction F1 Δ = -0.067 (95% CI [-0.068, -0.066], d_z = -17.50, paired t p = <1e-6, Holm p = <1e-6, n = 30)



corruption

Cost result

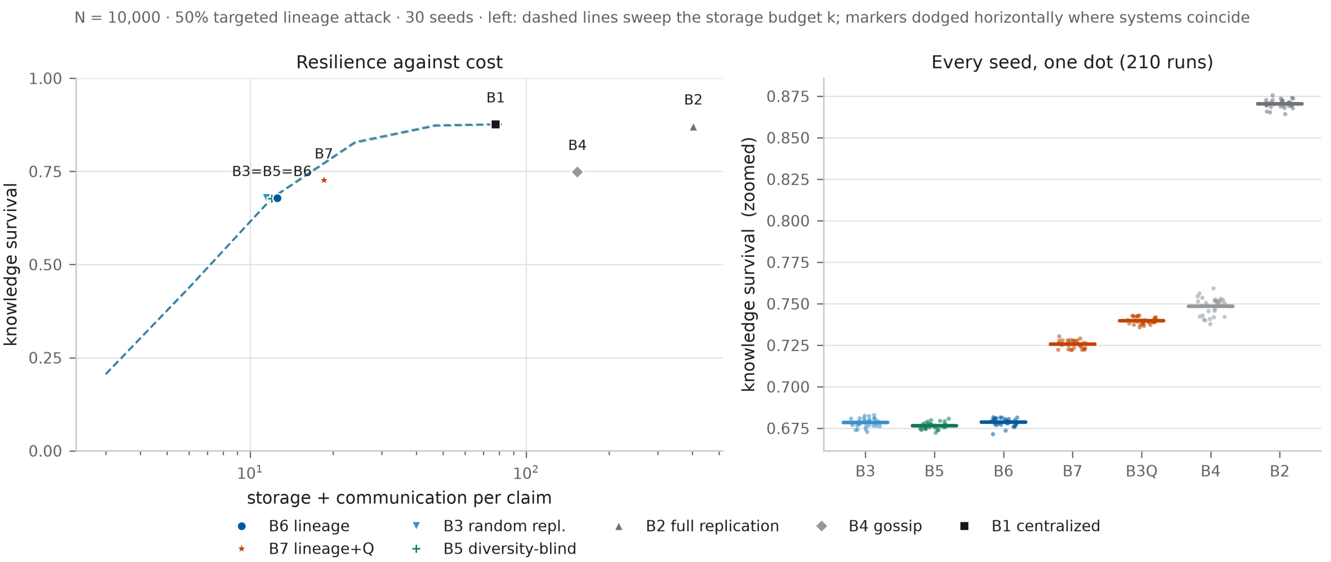
Storage is counted as published replicas per claim; communication counts every message: publication, questioning, and every probe attempt including attempts on dead hosts.

sys	KS	storage/claim	msgs/claim	redundancy eff.	comm. eff.
B0	0.1106	0.0000	0.0000	--	--
B1	0.8761	36.3279	41.2864	0.0241	0.0212
B2	0.8704	193.7486	210.0891	0.0045	0.0041
B3	0.6784	4.0000	8.0000	0.1696	0.0848
B4	0.7486	48.4371	105.0899	0.0155	0.0071
B5	0.6766	4.0000	8.0000	0.1692	0.0846
B6	0.6787	4.0000	8.0000	0.1697	0.0848

sys	KS	storage/claim	msgs/claim	redundancy eff.	comm. eff.
B7	0.7257	4.0000	14.5165	0.1814	0.0500

Sweeping the storage budget k ∈ {1, 2, 4, 8, 16, 32} traces each equal-budget policy's own resilience/cost curve, so the frontier claim does not depend on one arbitrary budget:

sys	k	storage/claim	msgs/claim	KS	accuracy	ISS
B3	1	1.000	2.000	0.205	0.115	0.133
B3	2	2.000	4.000	0.435	0.261	0.297
B3	4	4.000	8.000	0.679	0.442	0.504
B3	8	8.000	16.000	0.829	0.567	0.689
B3	16	16.000	30.936	0.875	0.606	0.813
B3	32	32.000	49.350	0.877	0.606	0.892
B5	1	1.000	2.000	0.207	0.113	0.133
B5	2	2.000	4.000	0.438	0.256	0.283
B5	4	4.000	8.000	0.677	0.430	0.437
B5	8	8.000	16.000	0.826	0.557	0.533
B5	16	16.000	30.976	0.872	0.598	0.563
B5	32	32.000	49.691	0.876	0.602	0.566
B6	1	1.000	2.000	0.205	0.119	0.133
B6	2	2.000	4.000	0.438	0.270	0.315
B6	4	4.000	8.000	0.679	0.452	0.550
B6	8	8.000	16.000	0.829	0.578	0.773
B6	16	16.000	30.961	0.873	0.616	0.918
B6	32	32.000	49.618	0.876	0.618	0.982



frontier

Negative results

Every condition in which the proposed architecture is **significantly worse** than a baseline on a primary metric (paired t-test, $p < 0.05$, uncorrected — deliberately the permissive threshold, so that nothing is hidden by a correction) is enumerated in `tables/table8_negative_results.csv`. There are 1295 such (condition, metric, comparison) entries.

system	loses to	metric	n_conditions	worst condition	worst gap
B6	B2	iss	161	targeted_whitebox sev=0.9 none@0.0 N=10000	-0.712
B6	B4	iss	161	random sev=0.7 none@0.0 N=100000	-0.353
B7	B2	iss	161	targeted_whitebox sev=0.9 none@0.0 N=10000	-0.712
B7	B3Q	accuracy_macro	152	partition sev=0.0 none@0.0 N=1000	-0.100
B6	B2	knowledge_survival	135	targeted_whitebox sev=0.9 none@0.0 N=10000	-0.640
B7	B3Q	knowledge_survival	126	partition sev=0.0 none@0.0 N=10000	-0.053
B6	B4	knowledge_survival	109	random sev=0.9 none@0.0 N=10000	-0.369

system	loses to	metric	n_conditions	worst condition	worst gap
B6	B2	accuracy_macro	89	targeted_whitebox sev=0.9 none@0.0 N=10000	-0.442
B6	B4	accuracy_macro	65	random sev=0.9 none@0.0 N=10000	-0.254
B7	B2	knowledge_survival	65	targeted_whitebox sev=0.9 none@0.0 N=1000	-0.631
B7	B2	accuracy_macro	36	targeted_whitebox sev=0.9 none@0.0 N=1000	-0.419
B6	B3	knowledge_survival	12	targeted_whitebox sev=0.9 none@0.0 N=100	-0.012
B6	B5	knowledge_survival	10	targeted_whitebox sev=0.9 none@0.0 N=100	-0.012
B7	B6	knowledge_survival	7	random sev=0.3 node_coordinated@0.2 N=10000	-0.017
B7	B6	iss	2	random sev=0.5 none@0.0 N=100	-0.004
B6	B3	iss	2	targeted_whitebox sev=0.9 none@0.0 N=100	-0.006
B7	B3Q	iss	2	targeted_world sev=0.9 none@0.0 N=100	-0.009

Regimes where lineage awareness does not help

- **Single-origin claims.** ~35% of claims in the world have exactly one lineage root. No placement policy can create independent support that does not exist; per-subset accuracy for these claims is in `acc_single_root` in `tables/appendix_full_grid.csv`.
- **Raw availability at equal budget.** Under random churn, knowledge survival is set by how many replicas survive, not by whose evidence they are: the equal-budget policies B3, B5 and B6 are close, and the lineage advantage is confined to correctness, independent-support survival, contradiction detection and calibration.
- **Extreme churn.** As severity approaches 90%, every equal-budget policy converges toward the same floor; only systems that pay orders of magnitude more storage (B2, B4) stay high.
- **Questioning is not lineage-specific.** The G_Q table shows the same questioning machinery applied to a count-based system (B3+Q). Where the two gains are comparable, continual discovery is an orthogonal contribution rather than a consequence of lineage.
- **Questioning can amplify corruption.** Re-verification contacts the origin; if the origin is corrupt, it installs the false value. See the corruption rows in the G_Q table for the measured sign of the effect.
- **The aggregation rule is nearly inert.** The placement × aggregation factorial (`tables/table7_placement_x_aggregation.csv`) isolates what the lineage weighting itself buys once placement is held fixed.

Deliberately hostile worlds

Six variants of the world generator were run specifically to find where the architecture stops helping. Each keeps everything else fixed and changes one property of the world or of the questioning policy.

world variant	B6 acc	B6 – B3	p (B6–B3)	B7 – B6 (G_Q)	p (G_Q)	B3Q – B3
every claim has exactly one origin (no independent evidence exists)	0.546	0.002	0.0016	0.214	<1e-6	0.288
near-perfectly correlated evidence (heavier family tail, 25% of claims are 1-origin/100-copy)	0.571	0.014	<1e-6	0.190	<1e-6	0.252
75% of facts are revised	0.415	0.014	<1e-6	0.267	<1e-6	0.346
5% of facts are revised	0.650	0.013	<1e-6	0.149	<1e-6	0.194
8 randomly-targeted questions per claim (flooding)	0.576	0.013	<1e-6	0.320	<1e-6	0.331
50% of origin sources corrupted	0.576	0.013	<1e-6	0.189	<1e-6	0.244

Full per-variant results: `tables/table9_negative_regimes.csv`; paired tests: `tables/table9b_negative_regime_tests.csv`.

Placement × aggregation, standard world (50% white-box attack)

Note that for the lineage placement the two aggregation rules nearly coincide *by construction*: round-robin over origins publishes at most one replica per origin, so counting replicas and counting distinct origins give the same tally. The informative cells are the broad-replication ones (B2 vs B2L), where retrieved replicas greatly outnumber distinct origins.

cell	placement	aggregation	KS	accuracy	ISS	contr. F1	storage
B2	broad	count	0.871	0.597	0.993	0.896	194.036
B2L	broad	lineage	0.870	0.598	0.993	0.896	194.036
B3	random	count	0.679	0.442	0.504	0.351	4.000
B3L	random	lineage	0.679	0.442	0.504	0.347	4.000

cell	placement	aggregation	KS	accuracy	ISS	contr. F1	storage
B5	support-count	count	0.677	0.430	0.437	0.000	4.000
B6	lineage	lineage	0.679	0.452	0.550	0.509	4.000
B6C	lineage	count	0.679	0.451	0.549	0.509	4.000

Placement × aggregation, correlation-dominated world (50% white-box attack)

Same cells with a much heavier family tail (family_alpha 0.4) and 25% of claims in the 1-origin/100-copy class — the regime in which replica count and independent support diverge most.

cell	placement	aggregation	KS	accuracy	ISS	contr. F1	storage
B2	broad	count	0.868	0.583	0.989	0.833	875.100
B2L	broad	lineage	0.867	0.581	0.990	0.835	875.100
B3	random	count	0.679	0.437	0.559	0.323	4.000
B3L	random	lineage	0.679	0.438	0.558	0.316	4.000
B5	support-count	count	0.677	0.431	0.515	0.000	4.000
B6	lineage	lineage	0.679	0.446	0.602	0.507	4.000
B6C	lineage	count	0.678	0.446	0.601	0.509	4.000

Success criteria

The experiment was specified with three criteria. They are evaluated here against the measured numbers, independently of whether the proposed architecture comes out ahead.

1. SCALE — does the behaviour persist at N ≥ 10■? MET

N	B6–B3 accuracy	Holm p	B6–B3 ISS	B7–B6 accuracy (G_Q)	seeds
100	0.008	0.00082	0.052	0.196	30
1,000	0.010	0.0023	0.047	0.185	30
10,000	0.010	<1e-6	0.046	0.189	30
100,000	0.009	<1e-6	0.046	0.187	10

2. RESILIENCE — KS(lineage) > KS(replication) at comparable budget under correlated or targeted failure? PARTIALLY MET

condition	comparison	ΔKS	d_z	Holm p	significant
50% loss, whole organisations	B6 – B3	0.002	0.908	8.1e-05	yes
50% loss, whole organisations	B6 – B5	0.002	0.755	0.00056	yes
70% loss, whole failure domains	B6 – B3	0.001	0.333	0.16	no
70% loss, whole failure domains	B6 – B5	0.002	0.446	0.062	no
50% targeted lineage attack	B6 – B3	0.000	0.100	0.59	no
50% targeted lineage attack	B6 – B5	0.002	0.774	0.00041	yes

3 of 6 equal-budget survival comparisons are positive and significant. Where they are not, the correct reading is that at matched storage the three policies keep roughly the same *amount* of knowledge alive; what differs is whether what survives is independent, and whether the answer built from it is correct.

3. DISCOVERY — does continual questioning produce a statistically meaningful improvement? MET

condition	metric	G_Q	d_z	Holm p	significant
50% loss, whole organisations	stale-fact accuracy	0.451	44.437	<1e-6	yes
50% loss, whole organisations	contradiction resolution	0.155	12.246	<1e-6	yes
50% loss, whole organisations	downstream accuracy	0.189	37.157	<1e-6	yes
50% targeted lineage attack	stale-fact accuracy	0.349	48.481	<1e-6	yes
50% targeted lineage attack	contradiction resolution	0.118	9.541	<1e-6	yes
50% targeted lineage attack	downstream accuracy	0.143	32.914	<1e-6	yes
20% origin-source corruption	stale-fact accuracy	0.395	29.488	<1e-6	yes
20% origin-source corruption	contradiction resolution	0.049	6.494	<1e-6	yes
20% origin-source corruption	downstream accuracy	0.165	29.336	<1e-6	yes
4-way partition, after reconnect	stale-fact accuracy	0.536	73.862	<1e-6	yes

condition	metric	G_Q	d_z	Holm p	significant
4-way partition, after reconnect	contradiction resolution	0.249	16.428	<1e-6	yes
4-way partition, after reconnect	downstream accuracy	0.314	49.747	<1e-6	yes

Important qualification. The identical questioning machinery applied to a count-based system gains +0.245 accuracy, which is larger than the +0.189 it gains on top of lineage. Criterion 3 is therefore met for continual discovery as a mechanism, but the data do **not** support the stronger claim that questioning depends on, or is amplified by, lineage awareness.

Statistical tests

1953 paired comparisons are recorded in `tables/table2_statistical_tests.csv`. Every comparison is paired by seed — the same world, workload and failure draw is given to both systems — and reports the mean difference, a bootstrap 95% CI of that difference (10,000 resamples), Cohen's d_z, a paired t-test, a Wilcoxon signed-rank test, and a Holm–Bonferroni corrected p-value computed within each metric's comparison family.

condition	comparison	metric	mean_a	mean_b	diff	ci	d_z	p_t	p_wilcoxon	p_holm	sig
50% targeted lineage attack	B6 – B3	knowledge_survival	0.679	0.678	0.000	[-0.001, 0.001]	0.100	0.59	0.65	0.59	no
50% targeted lineage attack	B6 – B3	accuracy_macro	0.451	0.441	0.010	[0.009, 0.012]	2.288	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B6 – B3	iss	0.550	0.504	0.046	[0.046, 0.047]	21.303	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B6 – B5	knowledge_survival	0.679	0.677	0.002	[0.001, 0.003]	0.774	0.00021	0.00086	0.00041	yes
50% targeted lineage attack	B6 – B5	accuracy_macro	0.451	0.429	0.022	[0.021, 0.023]	6.223	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B6 – B5	iss	0.550	0.437	0.113	[0.112, 0.114]	47.212	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B6 – B2	knowledge_survival	0.679	0.870	-0.192	[-0.192, -0.191]	-96.579	<1e-6	1.7e-06	<1e-6	yes
50% targeted lineage attack	B6 – B2	accuracy_macro	0.451	0.597	-0.146	[-0.146, -0.145]	-54.501	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B6 – B2	iss	0.550	0.993	-0.443	[-0.444, -0.442]	-194.644	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B7 – B6	knowledge_survival	0.726	0.679	0.047	[0.046, 0.048]	16.405	<1e-6	1.7e-06	<1e-6	yes
50% targeted lineage attack	B7 – B6	accuracy_macro	0.594	0.451	0.143	[0.141, 0.144]	32.914	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B7 – B6	iss	0.550	0.550	-0.000	[-0.001, 0.000]	-0.136	0.46	0.33	0.46	no
50% targeted lineage attack	B7 – B3Q	knowledge_survival	0.726	0.740	-0.014	[-0.015, -0.013]	-4.766	<1e-6	1.7e-06	<1e-6	yes
50% targeted lineage attack	B7 – B3Q	accuracy_macro	0.594	0.628	-0.034	[-0.036, -0.032]	-5.931	<1e-6	<1e-6	<1e-6	yes
50% targeted lineage attack	B7 – B3Q	iss	0.550	0.507	0.043	[0.042, 0.043]	19.802	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B6 – B3	knowledge_survival	0.822	0.820	0.002	[0.001, 0.003]	0.908	2.7e-05	0.00015	8.1e-05	yes
50% loss, whole organisations	B6 – B3	accuracy_macro	0.575	0.561	0.014	[0.013, 0.016]	3.906	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B6 – B3	iss	0.687	0.622	0.065	[0.064, 0.066]	24.259	<1e-6	<1e-6	<1e-6	yes

condition	comparison	metric	mean_a	mean_b	diff	ci	d_z	p_t	p_wilcoxon	p_holm	sig
50% loss, whole organisations	B6 – B5	knowledge_survival	0.822	0.820	0.002	[0.001, 0.003]	0.755	0.00028	0.00072	0.00056	yes
50% loss, whole organisations	B6 – B5	accuracy_macro	0.575	0.550	0.025	[0.023, 0.026]	5.937	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B6 – B5	iss	0.687	0.530	0.157	[0.156, 0.158]	60.695	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B6 – B2	knowledge_survival	0.822	0.877	-0.054	[-0.055, -0.053]	-22.602	<1e-6	1.7e-06	<1e-6	yes
50% loss, whole organisations	B6 – B2	accuracy_macro	0.575	0.605	-0.030	[-0.031, -0.029]	-8.872	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B6 – B2	iss	0.687	1.000	-0.313	[-0.314, -0.312]	-116.311	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B7 – B6	knowledge_survival	0.883	0.822	0.061	[0.060, 0.062]	18.591	<1e-6	1.7e-06	<1e-6	yes
50% loss, whole organisations	B7 – B6	accuracy_macro	0.764	0.575	0.189	[0.187, 0.190]	37.157	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B7 – B6	iss	0.687	0.687	0.001	[-0.000, 0.002]	0.202	0.28	0.33	0.28	no
50% loss, whole organisations	B7 – B3Q	knowledge_survival	0.883	0.898	-0.015	[-0.016, -0.015]	-6.462	<1e-6	1.7e-06	<1e-6	yes
50% loss, whole organisations	B7 – B3Q	accuracy_macro	0.764	0.806	-0.042	[-0.043, -0.040]	-9.171	<1e-6	<1e-6	<1e-6	yes
50% loss, whole organisations	B7 – B3Q	iss	0.687	0.628	0.059	[0.058, 0.060]	25.209	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B3	knowledge_survival	0.667	0.665	0.001	[-0.000, 0.003]	0.333	0.078	0.1	0.16	no
70% loss, whole failure domains	B6 – B3	accuracy_macro	0.442	0.431	0.012	[0.010, 0.013]	2.503	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B3	iss	0.499	0.471	0.028	[0.027, 0.029]	8.576	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B5	knowledge_survival	0.667	0.665	0.002	[0.000, 0.003]	0.446	0.021	0.025	0.062	no
70% loss, whole failure domains	B6 – B5	accuracy_macro	0.442	0.421	0.021	[0.020, 0.023]	4.622	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B5	iss	0.499	0.430	0.069	[0.068, 0.070]	21.502	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B2	knowledge_survival	0.667	0.876	-0.209	[-0.210, -0.208]	-69.205	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B2	accuracy_macro	0.442	0.604	-0.162	[-0.163, -0.161]	-44.999	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B6 – B2	iss	0.499	0.998	-0.499	[-0.501, -0.498]	-132.303	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B7 – B6	knowledge_survival	0.716	0.667	0.049	[0.048, 0.051]	12.339	<1e-6	1.7e-06	<1e-6	yes
70% loss, whole failure domains	B7 – B6	accuracy_macro	0.586	0.442	0.144	[0.142, 0.146]	29.140	<1e-6	<1e-6	<1e-6	yes
70% loss, whole failure domains	B7 – B6	iss	0.499	0.499	0.000	[-0.001, 0.002]	0.131	0.48	0.52	0.48	no
70% loss, whole failure domains	B7 – B3Q	knowledge_survival	0.716	0.727	-0.011	[-0.012, -0.010]	-2.657	<1e-6	1.7e-06	<1e-6	yes
70% loss, whole failure domains	B7 – B3Q	accuracy_macro	0.586	0.617	-0.031	[-0.033, -0.029]	-5.217	<1e-6	<1e-6	<1e-6	yes

condition	comparison	metric	mean_a	mean_b	diff	ci	d_z	p_t	p_wilcoxon	p_holm	sig
70% loss, whole failure domains	B7 – B3Q	iss	0.499	0.473	0.026	[0.025, 0.027]	8.160	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B3	knowledge_survival	0.717	0.702	0.014	[0.013, 0.015]	5.128	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B3	accuracy_macro	0.475	0.456	0.019	[0.018, 0.020]	5.142	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B3	iss	0.810	0.706	0.104	[0.103, 0.105]	53.870	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B5	knowledge_survival	0.717	0.695	0.022	[0.021, 0.023]	6.418	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B5	accuracy_macro	0.475	0.444	0.031	[0.029, 0.032]	7.136	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B5	iss	0.810	0.561	0.249	[0.248, 0.250]	104.339	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B2	knowledge_survival	0.717	0.719	-0.003	[-0.004, -0.002]	-0.962	1.2e-05	6.9e-05	4.8e-05	yes
20% origin-source corruption	B6 – B2	accuracy_macro	0.475	0.464	0.011	[0.010, 0.013]	2.872	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B6 – B2	iss	0.810	1.000	-0.190	[-0.191, -0.189]	-91.736	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B7 – B6	knowledge_survival	0.770	0.717	0.054	[0.053, 0.055]	15.508	<1e-6	1.7e-06	<1e-6	yes
20% origin-source corruption	B7 – B6	accuracy_macro	0.640	0.475	0.165	[0.163, 0.167]	29.336	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B7 – B6	iss	0.810	0.810	-0.000	[-0.001, 0.000]	-0.227	0.22	0.19	0.22	no
20% origin-source corruption	B7 – B3Q	knowledge_survival	0.770	0.770	-0.000	[-0.001, 0.001]	-0.029	0.87	0.99	0.87	no
20% origin-source corruption	B7 – B3Q	accuracy_macro	0.640	0.667	-0.027	[-0.028, -0.025]	-6.903	<1e-6	<1e-6	<1e-6	yes
20% origin-source corruption	B7 – B3Q	iss	0.810	0.715	0.095	[0.095, 0.096]	61.064	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B3	knowledge_survival	0.821	0.820	0.000	[-0.001, 0.001]	0.160	0.39	0.37	0.4	no
50% random churn	B6 – B3	accuracy_macro	0.574	0.562	0.012	[0.010, 0.014]	2.780	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B3	iss	0.686	0.622	0.064	[0.063, 0.065]	23.867	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B5	knowledge_survival	0.821	0.820	0.001	[-0.000, 0.002]	0.282	0.13	0.17	0.4	no
50% random churn	B6 – B5	accuracy_macro	0.574	0.551	0.023	[0.022, 0.024]	6.243	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B5	iss	0.686	0.530	0.156	[0.155, 0.157]	57.753	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B2	knowledge_survival	0.821	0.876	-0.055	[-0.056, -0.055]	-24.182	<1e-6	1.7e-06	<1e-6	yes
50% random churn	B6 – B2	accuracy_macro	0.574	0.605	-0.031	[-0.032, -0.030]	-8.500	<1e-6	<1e-6	<1e-6	yes
50% random churn	B6 – B2	iss	0.686	1.000	-0.314	[-0.315, -0.313]	-104.606	<1e-6	<1e-6	<1e-6	yes
50% random churn	B7 – B6	knowledge_survival	0.881	0.821	0.060	[0.059, 0.061]	23.198	<1e-6	1.7e-06	<1e-6	yes

condition	comparison	metric	mean_a	mean_b	diff	ci	d_z	p_t	p_wilcoxon	p_holm	sig
50% random churn	B7 – B6	accuracy_macro	0.763	0.574	0.189	[0.188, 0.191]	44.472	<1e-6	<1e-6	<1e-6	yes
50% random churn	B7 – B6	iss	0.686	0.686	0.000	[-0.001, 0.001]	0.031	0.87	0.56	0.87	no
50% random churn	B7 – B3Q	knowledge_survival	0.881	0.899	-0.017	[-0.018, -0.016]	-7.040	<1e-6	1.7e-06	<1e-6	yes
50% random churn	B7 – B3Q	accuracy_macro	0.763	0.807	-0.044	[-0.046, -0.043]	-9.081	<1e-6	<1e-6	<1e-6	yes
50% random churn	B7 – B3Q	iss	0.686	0.628	0.058	[0.057, 0.059]	23.178	<1e-6	<1e-6	<1e-6	yes
90% random churn	B6 – B3	knowledge_survival	0.300	0.301	-0.001	[-0.003, 0.000]	-0.253	0.18	0.24	0.53	no
90% random churn	B6 – B3	accuracy_macro	0.177	0.174	0.003	[0.002, 0.004]	0.767	0.00023	0.00034	0.00046	yes
90% random churn	B6 – B3	iss	0.203	0.200	0.003	[0.001, 0.004]	0.769	0.00023	0.00021	0.0009	yes
90% random churn	B6 – B5	knowledge_survival	0.300	0.301	-0.001	[-0.003, 0.001]	-0.233	0.21	0.3	0.53	no
90% random churn	B6 – B5	accuracy_macro	0.177	0.169	0.008	[0.007, 0.009]	2.114	<1e-6	<1e-6	<1e-6	yes
90% random churn	B6 – B5	iss	0.203	0.195	0.008	[0.007, 0.010]	2.643	<1e-6	<1e-6	<1e-6	yes
90% random churn	B6 – B2	knowledge_survival	0.300	0.843	-0.543	[-0.544, -0.541]	-121.786	<1e-6	<1e-6	<1e-6	yes
90% random churn	B6 – B2	accuracy_macro	0.177	0.575	-0.398	[-0.399, -0.397]	-94.364	<1e-6	<1e-6	<1e-6	yes
90% random churn	B6 – B2	iss	0.203	0.895	-0.692	[-0.693, -0.691]	-254.301	<1e-6	<1e-6	<1e-6	yes
90% random churn	B7 – B6	knowledge_survival	0.322	0.300	0.022	[0.021, 0.023]	5.200	<1e-6	1.7e-06	<1e-6	yes
90% random churn	B7 – B6	accuracy_macro	0.237	0.177	0.060	[0.058, 0.062]	13.066	<1e-6	<1e-6	<1e-6	yes
90% random churn	B7 – B6	iss	0.203	0.203	0.000	[-0.001, 0.001]	0.067	0.71	0.76	1	no
90% random churn	B7 – B3Q	knowledge_survival	0.322	0.328	-0.006	[-0.007, -0.004]	-1.387	<1e-6	5.2e-06	<1e-6	yes
90% random churn	B7 – B3Q	accuracy_macro	0.237	0.251	-0.014	[-0.016, -0.012]	-2.433	<1e-6	<1e-6	<1e-6	yes
90% random churn	B7 – B3Q	iss	0.203	0.200	0.003	[0.002, 0.005]	1.069	2.4e-06	5.1e-06	1.2e-05	yes
4-way partition, after reconnect	B6 – B3	knowledge_survival	0.674	0.677	-0.002	[-0.003, -0.002]	-1.149	<1e-6	2.6e-05	2.1e-06	yes
4-way partition, after reconnect	B6 – B3	accuracy_macro	0.382	0.375	0.007	[0.005, 0.008]	1.715	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B6 – B3	iss	0.930	0.773	0.156	[0.156, 0.157]	90.835	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B6 – B5	knowledge_survival	0.674	0.676	-0.002	[-0.003, -0.001]	-0.747	0.00031	0.0012	0.00063	yes
4-way partition, after reconnect	B6 – B5	accuracy_macro	0.382	0.368	0.013	[0.012, 0.015]	3.043	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B6 – B5	iss	0.930	0.566	0.364	[0.363, 0.365]	162.473	<1e-6	<1e-6	<1e-6	yes

condition	comparison	metric	mean_a	mean_b	diff	ci	d_z	p_t	p_wilcoxon	p_holm	sig
4-way partition, after reconnect	B6 – B2	knowledge_survival	0.674	0.659	0.015	[0.013, 0.017]	2.748	<1e-6	1.9e-06	<1e-6	yes
4-way partition, after reconnect	B6 – B2	accuracy_macro	0.382	0.357	0.025	[0.023, 0.026]	5.809	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B6 – B2	iss	0.930	1.000	-0.070	[-0.071, -0.070]	-62.026	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B7 – B6	knowledge_survival	0.829	0.674	0.154	[0.151, 0.157]	20.884	<1e-6	1.7e-06	<1e-6	yes
4-way partition, after reconnect	B7 – B6	accuracy_macro	0.695	0.382	0.314	[0.311, 0.316]	49.747	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B7 – B6	iss	0.930	0.930	0.000	[0.000, 0.000]	0.000	1	1	1	no
4-way partition, after reconnect	B7 – B3Q	knowledge_survival	0.829	0.882	-0.053	[-0.054, -0.052]	-15.043	<1e-6	1.7e-06	<1e-6	yes
4-way partition, after reconnect	B7 – B3Q	accuracy_macro	0.695	0.792	-0.096	[-0.098, -0.095]	-19.265	<1e-6	<1e-6	<1e-6	yes
4-way partition, after reconnect	B7 – B3Q	iss	0.930	0.786	0.144	[0.144, 0.145]	92.506	<1e-6	<1e-6	<1e-6	yes

Limitations

- **No LLM calls anywhere.** Semantic reasoning is abstracted: a claim's value is a symbol and an answer is correct iff it equals the claim's current true value. Nothing here measures whether an agent can formulate a good question in natural language, extract evidence from text, or recognise that two differently-worded claims are the same claim. Every result is a claim about the *distribution layer*, not about end-to-end agents. This was both a scope decision (the hypothesis is about evidence topology under failure) and an environment constraint (no model credentials were available on the machine that ran this).
- **Simulation artefacts.** Placement, failure and retrieval are modelled, not deployed. The retrieval model assumes all live agents are mutually reachable outside the partition regime, which is a *conservative* assumption for the hypothesis: it maximises the value of having a replica anywhere at all, which favours the broad-replication baselines.
- **Re-verification is modelled as a privileged operation, and that inflates G_Q.** A REVERIFY question contacts the claim's origin, and the origin re-observes the world, so unless that origin is corrupt the refreshed value is correct by construction. Most of the measured questioning gain is therefore a measurement of *what re-verification is worth if you can do it*, at a stated message price — not evidence that formulating good questions is itself easy or cheap. The informative parts of the questioning ablation are the ones that are not tautological: that ADD_SUPPORT (publishing an additional independent origin — the genuinely lineage-flavoured question) contributes far less than re-verification, that returns saturate and then waste messages, that fragility-based targeting is not the best targeting policy, and that the same gain appears without lineage.
- **Contradiction ground truth is world-level**: a claim counts as contradictory if the evidence that exists anywhere in the world disagrees, including evidence no system stores. This is system-independent by design, but it caps the achievable recall of every small-budget policy and flatters broad replication, which probes far more of that evidence.
- **Recovery time is a logical-round proxy**, not wall-clock latency.
- **The white-box attacker is greedy and batched** (8 recomputation rounds), so it is a lower bound on an optimal adversary.
- **The scaling trends come from the measured scales only.** They are reported as empirical trends with log-log slopes and R^2, never as asymptotic complexity.
- **One world generator.** All results are conditional on the generative model of the world (heavy-tailed family sizes, ~35% single-origin claims, 25% revised facts, locality-correlated homes). The parameters are declared in `configs/` and can be swept.

Reproduction instructions

```
cd experiments/large_scale_agentic_web
pip install -r requirements.txt
./run_all.sh            # N = 1e2, 1e3, 1e4 + every ablation + analysis
FULL=1 ./run_all.sh     # additionally N = 1e5
QUICK=1 ./run_all.sh    # smoke run to verify the pipeline
```

Each invocation writes a new timestamped directory under `results/` containing the raw `rows.csv` and a `manifest.json` recording the git commit, the complete configuration, the seed list, the timestamp, the machine and every dependency version. Raw outputs are never overwritten. `results/index.jsonl` is an append-only index used by the analysis step; `results/summary.json` contains every number quoted above.

Seeds: `seeds.json` (fixed list). Superseded pilot runs are archived under `results/_pilot_pre_tiebreak_fix/` with a note explaining why they are not used.

Large-Scale Agentic Web Stress Test

Design. We stress-test lineage-first knowledge distribution on a simulated agentic web of N ∈ {10^2, 10^3, 10^4, 10^5} agents organised into teams, failure domains, organisations and regions. Knowledge is partial and local: every evidence item has a home agent, and each item descends from exactly one origin observation. Family sizes are heavy-tailed and ~35% of claims have a single origin, so replica count and independent evidential support come apart by construction. We evaluate 127,374 labelled questions per seed (single-hop, multi-hop, temporal, revision-sensitive, reconstruction, contradiction, evidence-verification, strategic ranking) against eight systems: isolated memory, a centralized store, full replication, random replication, gossip, a diversity-blind replica-count policy, the lineage-aware fabric, and the fabric with continual questioning. Equal-budget systems publish the identical number of replicas per claim (4); every probe attempt is charged as a message, including attempts on dead hosts. Failures escalate from 10–90% random churn to correlated loss of whole failure domains, organisations and regions, to a **white-box adversary** that knows each system's own placement and greedily removes the hosts of its scarcest surviving independent evidence, to partitions with independent updates, to corruption of up to 20% of nodes or of origin sources. 30 seeds, paired by seed, Holm-corrected.

Result. Under the white-box attack at N = 10,000 removing half the agents, knowledge survival is statistically indistinguishable across equal-budget policies (0.679 lineage, 0.678 random, 0.677 diversity-blind), but the lineage fabric is better on every provenance-sensitive quantity: task accuracy +0.010 (d_z = 2.29, Holm p <1e-6), independent-support survival +0.046 (d_z = 21.30), and contradiction F1 +0.155. The gap widens under origin-source corruption, where a corrupted origin's false value is inherited by all of its copies: accuracy +0.019 (Holm p <1e-6). Full replication survives better (0.870) at 194 replicas per claim — 48× the storage. The frontier (Fig. 2) therefore reads: at matched budget the three equal-budget policies coincide on survival and separate only on evidential quality, and the margin that broad replication holds over all of them is bought with two orders of magnitude more storage rather than with a better use of it.

Questioning ablation. At *identical storage* — the base budget is reduced by exactly the replicas questioning adds — continual questioning raises accuracy by +0.189 (d_z = 37.16, Holm p <1e-6) for a 1.81× increase in messages, and drives post-partition reconciliation (+0.314 accuracy after reconnection).

Negative findings. The same questioning machinery on a *count-based* system gains +0.245: continual discovery is largely orthogonal to lineage, not a consequence of it. Under corruption, re-verification can pull the false value from a corrupt origin (+0.165 accuracy). Lineage awareness does not improve raw survival under random churn (+0.000), cannot help the ~35% of claims with a single origin, and the placement × aggregation factorial shows that essentially all of the benefit comes from *which evidence is published*, not from lineage-weighted voting.

Table 1 (main results)

Table 1. Knowledge survival and downstream accuracy under four failure conditions at N = 10,000 (30 seeds, paired). KS is the fraction of valid knowledge that is both recoverable and answered correctly; ISS is the fraction of original independent evidence paths that survive in the system's own stored set; storage counts published replicas per claim and messages count every publication, question and probe attempt. B3, B5, B6 and B7 share an identical storage budget; B1, B2 and B4 do not, and their cost columns show what their margin is bought with. 95% intervals and all remaining conditions are in the appendix.

failure condition	system	KS	accuracy	ISS	contr. F1	storage/claim	msgs/claim
50% random churn	B1	0.730	0.499	0.833	0.781	36.328	42.475
50% random churn	B2	0.876	0.605	1.000	0.900	193.749	209.462
50% random churn	B3	0.820	0.562	0.622	0.459	4.000	8.000
50% random churn	B4	0.864	0.586	0.960	0.896	48.437	99.588
50% random churn	B5	0.820	0.551	0.530	0.000	4.000	8.000
50% random churn	B6	0.821	0.574	0.686	0.638	4.000	8.000
50% random churn	B7	0.881	0.763	0.686	0.486	4.000	14.517
50% loss, whole organisations	B1	0.789	0.539	0.900	0.843	36.328	41.998
50% loss, whole organisations	B2	0.877	0.605	1.000	0.899	193.749	209.472
50% loss, whole organisations	B3	0.820	0.561	0.622	0.462	4.000	8.000
50% loss, whole organisations	B4	0.787	0.494	0.750	0.772	48.437	102.565
50% loss, whole organisations	B5	0.820	0.550	0.530	0.000	4.000	8.000
50% loss, whole organisations	B6	0.822	0.575	0.687	0.639	4.000	8.000
50% loss, whole organisations	B7	0.883	0.764	0.687	0.489	4.000	14.517
50% targeted lineage attack	B1	0.876	0.600	1.000	0.937	36.328	41.286
50% targeted lineage attack	B2	0.870	0.597	0.993	0.895	193.749	210.089
50% targeted lineage attack	B3	0.678	0.441	0.504	0.351	4.000	8.000

failure condition	system	KS	accuracy	ISS	contr. F1	storage/claim	msgs/claim
50% targeted lineage attack	B4	0.749	0.472	0.691	0.747	48.437	105.090
50% targeted lineage attack	B5	0.677	0.429	0.437	0.000	4.000	8.000
50% targeted lineage attack	B6	0.679	0.451	0.550	0.506	4.000	8.000
50% targeted lineage attack	B7	0.726	0.594	0.550	0.398	4.000	14.517
20% origin-source corruption	B1	0.675	0.431	0.933	0.870	36.328	41.763
20% origin-source corruption	B2	0.719	0.464	1.000	0.894	193.749	205.168
20% origin-source corruption	B3	0.702	0.456	0.706	0.628	4.000	8.000
20% origin-source corruption	B4	0.719	0.461	0.994	0.901	48.437	96.845
20% origin-source corruption	B5	0.695	0.444	0.561	0.000	4.000	8.000
20% origin-source corruption	B6	0.717	0.475	0.810	0.827	4.000	8.000
20% origin-source corruption	B7	0.770	0.640	0.810	0.794	4.000	14.517

Figure captions

Figure 1. Knowledge survival and task accuracy as failure escalates (N = 10,000, 30 seeds, bands are 95% CIs). Columns are three failure structures at matched node-level severity: uniform random churn, correlated loss of whole organisations, and a white-box adversary that knows each system's placement and removes the hosts of its scarcest surviving independent evidence first. Axes span the full [0, 1] range. The equal-budget policies (B3, B5, B6) separate on accuracy rather than on availability; the systems that stay flat at extreme severity (B2, B4) do so at one to two orders of magnitude more storage, quantified in Figure 2.

Figure 2. Resilience–efficiency frontier under the 50% white-box attack (N = 10,000). Points are the eight systems; dashed lines sweep the storage budget $k \in \{1, 2, 4, 8, 16, 32\}$ for the three equal-budget policies, so the comparison does not depend on one arbitrary budget. Left: knowledge survival against storage cost. Right: against storage plus communication. Error bars are 95% CIs; the x-axis is logarithmic because the systems differ by two orders of magnitude in cost.

Appendix table

Table A1. Storage-budget sweep under the 50% white-box attack (N = 10,000, 15 seeds). Each equal-budget policy is run at $k \in \{1, 2, 4, 8, 16, 32\}$ published replicas per claim. Knowledge survival converges as k grows — availability is set by replica count — while independent-support survival does not: the diversity-blind policy saturates well below the lineage policy at every budget, because additional replicas of one origin add no independent support.

system	budget k	storage/claim	msgs/claim	KS	accuracy	ISS
B3	1	1.000	2.000	0.205	0.115	0.133
B3	2	2.000	4.000	0.435	0.261	0.297
B3	4	4.000	8.000	0.679	0.442	0.504
B3	8	8.000	16.000	0.829	0.567	0.689
B3	16	16.000	30.936	0.875	0.606	0.813
B3	32	32.000	49.350	0.877	0.606	0.892
B5	1	1.000	2.000	0.207	0.113	0.133
B5	2	2.000	4.000	0.438	0.256	0.283
B5	4	4.000	8.000	0.677	0.430	0.437
B5	8	8.000	16.000	0.826	0.557	0.533
B5	16	16.000	30.976	0.872	0.598	0.563
B5	32	32.000	49.691	0.876	0.602	0.566
B6	1	1.000	2.000	0.205	0.119	0.133
B6	2	2.000	4.000	0.438	0.270	0.315
B6	4	4.000	8.000	0.679	0.452	0.550
B6	8	8.000	16.000	0.829	0.578	0.773
B6	16	16.000	30.961	0.873	0.616	0.918
B6	32	32.000	49.618	0.876	0.618	0.982